

KNUST RANKS NO.1 GLOBALLY FOR THE
PROVISION OF QUALITY EDUCATION (SDG 4)



Discrete-Time Signals and Systems

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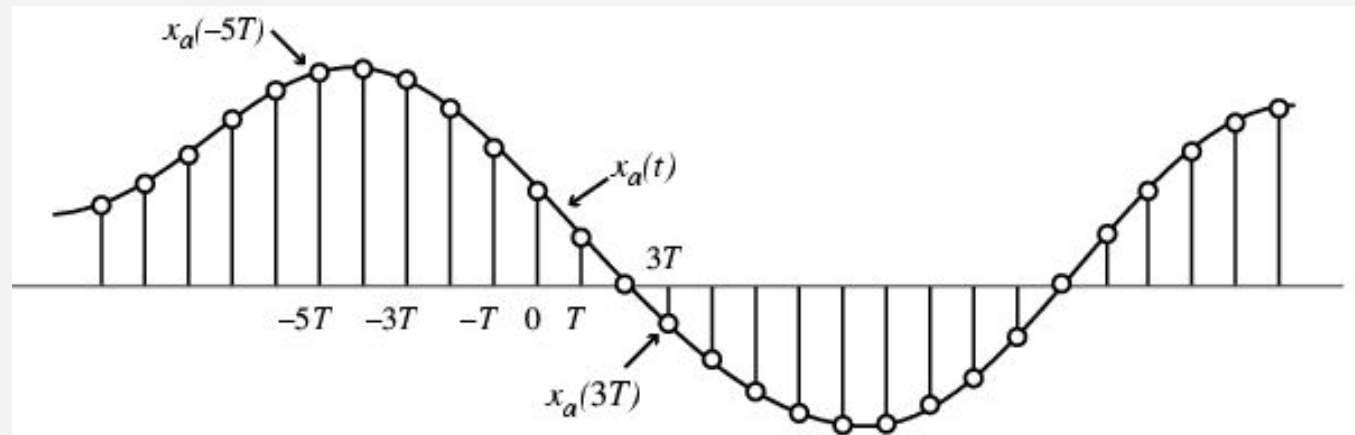


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Discrete-Time Signals: Time-Domain Representation

Signals are represented as a sequences of numbers, called **samples**

In some applications, a discrete-time sequence $\{x[n]\}$ may be generated by periodically sampling a continuous-time signal $x_a(t)$ at uniform intervals of time



Discrete-Time Signals: Time-Domain Representation

- Here, the n-th sample is given by

$$x[n] = x_a(t) \big|_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, \dots$$

- The spacing T between two consecutive samples is called the **sampling interval** or **sampling period**
- Reciprocal of sampling interval T , denoted as F_T , is called the **sampling frequency**:

$$F_T = 1/T$$



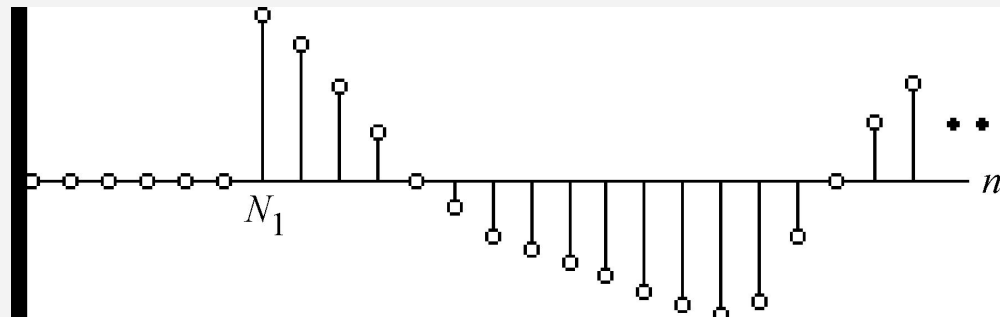
Discrete-Time Signals: Time-Domain Representation

- Two types of discrete-time signals:
 - **Sampled-data signals** in which samples are continuous-valued
 - **Digital signals** in which samples are discrete-valued
- Signals in a practical digital signal processing system are digital signals obtained by quantizing the sample values either by rounding or truncation



Discrete-Time Signals: Time-Domain Representation

- A right-sided sequence $x[n]$ has zero-valued samples for $n < N_1$



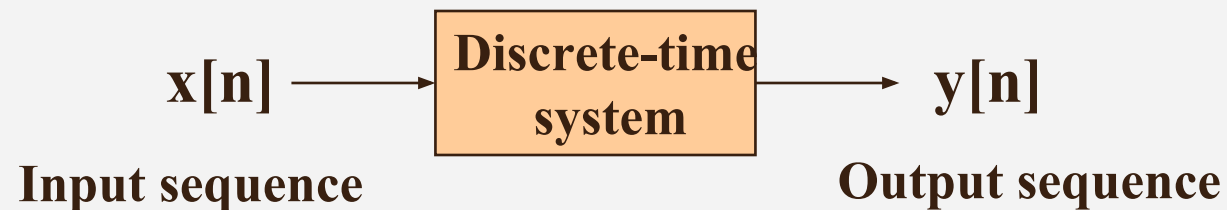
A right-sided sequence

- If $N_1 \geq 0$, a right-sided sequence is called a **causal sequence**



Operations on Sequences

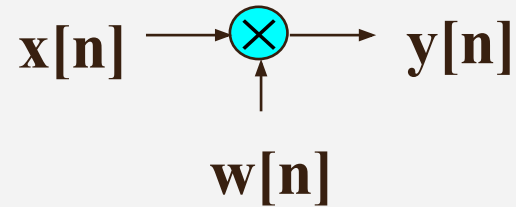
- A single-input, single-output discrete-time system operates on a sequence, called the **input sequence**, according to some prescribed rules and develops another sequence, called the **output sequence**, with more desirable properties



Basic Operations

- Product (modulation) operation:

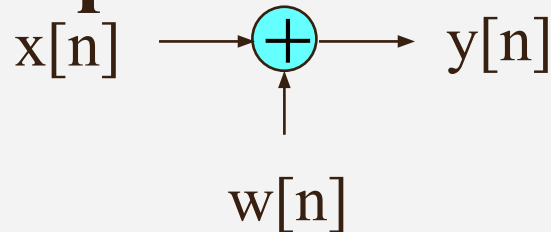
Modulator



$$y[n] = \mathbf{x[n] \cdot w[n]}$$

- Addition operation:

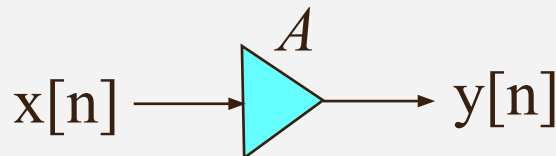
Adder



$$y[n] = \mathbf{x[n] + w[n]}$$

- Scalar Multiplication operation:

Multiplier



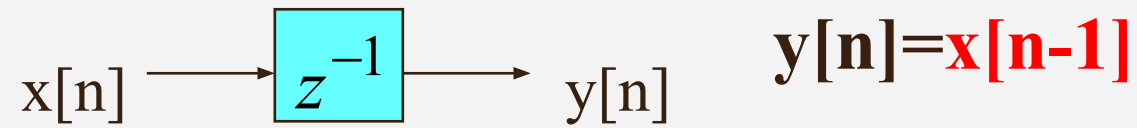
$$y[n] = \mathbf{A \cdot x[n]}$$



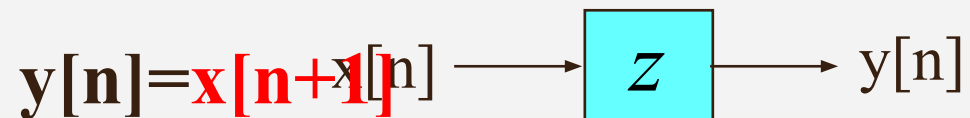
Basic Operations

▪ Time-shifting operation:

–Unit delay



–Unit advance

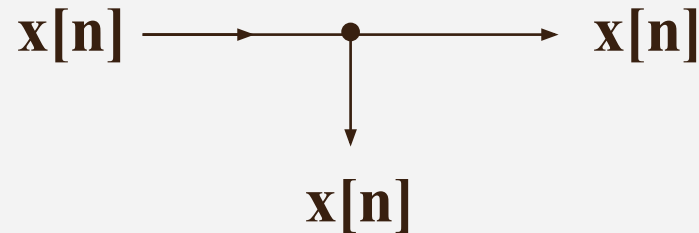


Basic Operations

- Time-reversal (folding) operation:

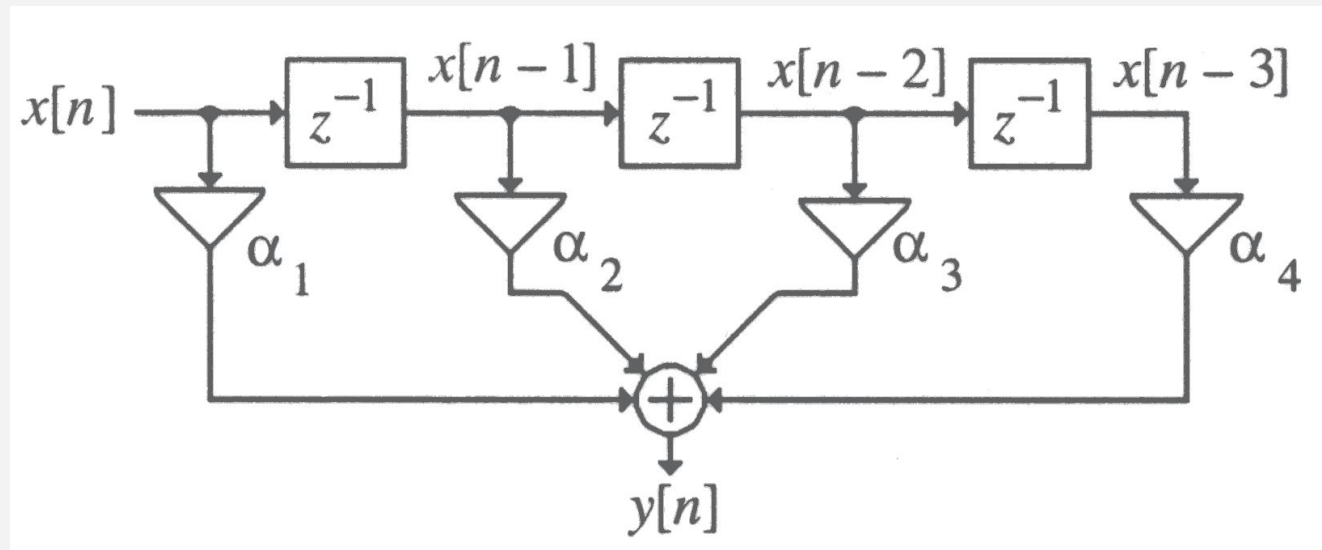
$$y[n] = x[-n]$$

- Branching operation: Used to provide multiple copies of a sequence



Basic Operations

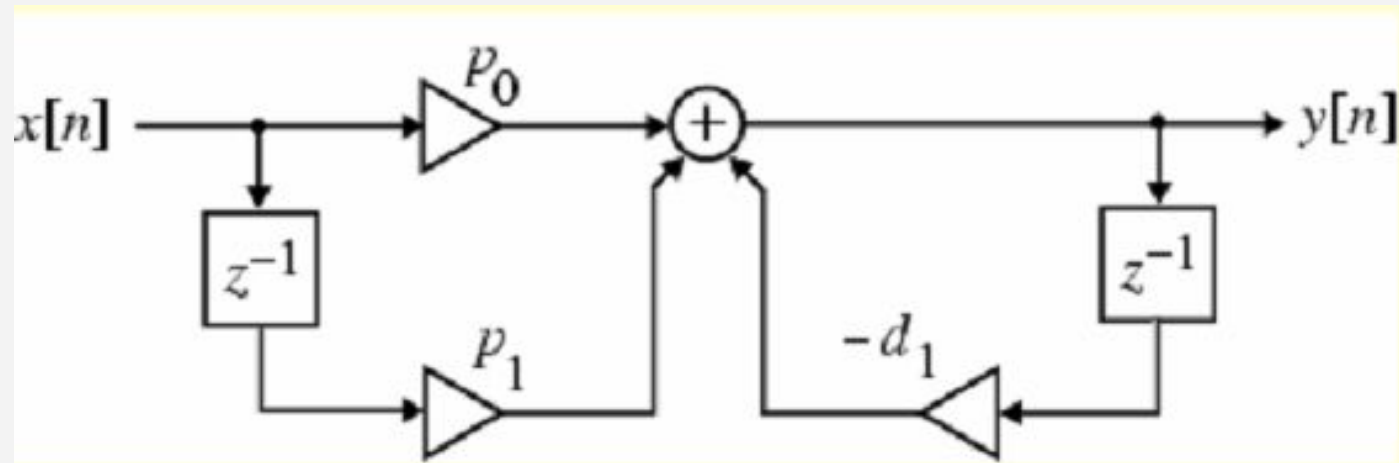
▪ Example -



$$y[n] = \alpha_1 x[n] + \alpha_2 x[n-1] + \alpha_3 x[n-2] + \alpha_4 x[n-3]$$



Combination of Elementary Operations



Sampling Rate Alteration

- Employed to generate a new sequence $y[n]$ with a sampling rate F'_T higher or lower than that of the sampling rate F_T of a given sequence $x[n]$
- Sampling rate alteration ratio is

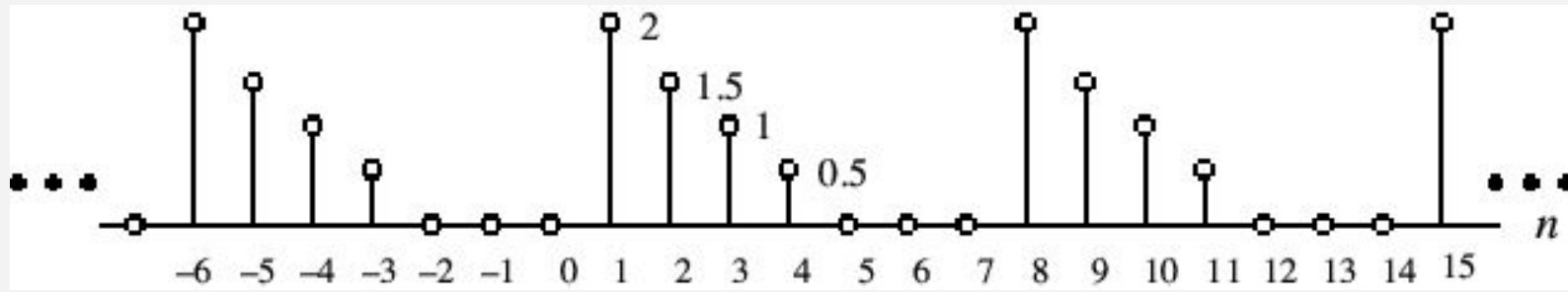
$$R = F'_T / F_T$$

- If $R > 1$, the process called **interpolation**
- If $R < 1$, the process called **decimation**



Classification of Sequences based on periodicity

- Example -



- A sequence satisfying the periodicity condition is called a **periodic sequence**



Classification of Sequences Energy and Power Signals

- Total energy of a sequence $x[n]$ is defined by

$$\mathcal{E}_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

- An infinite length sequence with finite sample values may or may not have finite energy
- A finite length sequence with finite sample values has finite energy



Classification of Sequences Energy and Power Signals

- The average power of an aperiodic sequence is defined by

$$P_x = \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x[n]|^2$$

- Define the energy of a sequence $x[n]$ over a finite interval $-K \leq n \leq K$ as

$$\mathcal{E}_{x,K} = \sum_{n=-K}^K |x[n]|^2$$



Classification of Sequences Energy and Power Signals

- An infinite energy signal with finite average power is called a **power signal**

Example - A periodic sequence which has a finite average power but infinite energy

- A finite energy signal with zero average power is called an **energy signal**

Example - A finite-length sequence which has finite energy but zero average power



Classification of Sequences bounded, absolutely summable and square summable

- A sequence $x[n]$ is said to be **bounded** if

$$|x[n]| \leq B_x < \infty$$

- Example - The sequence $x[n] = \cos(0.3\pi n)$ is a bounded sequence as

$$|x[n]| = |\cos 0.3\pi n| \leq 1$$



Classification of Sequences bounded, absolutely summable and square summable

- A sequence $x[n]$ is said to be **absolutely summable** if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- Example - The sequence

$$y[n] = \begin{cases} 0.3^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

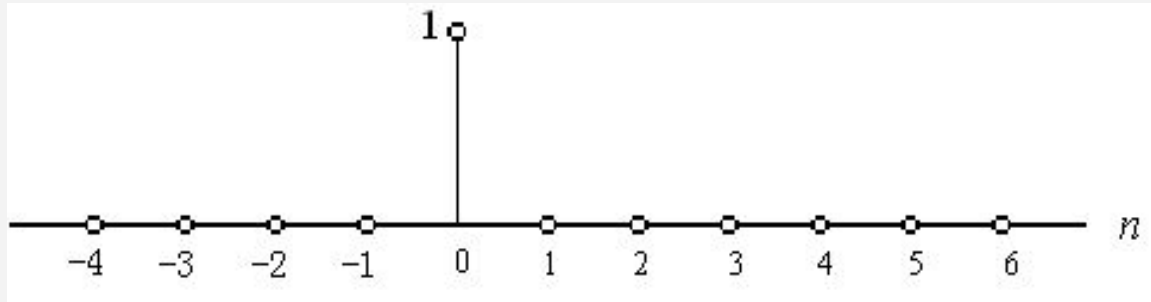
is an absolutely summable sequence as

$$\sum_{n=0}^{\infty} |0.3^n| = \frac{1}{1-0.3} = 1.42857 < \infty$$



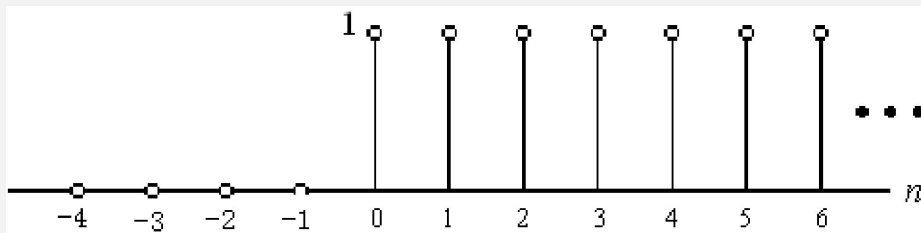
Basic Sequences

■ Unit sample sequence -



$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

• Unit step sequence -



$$\mu[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



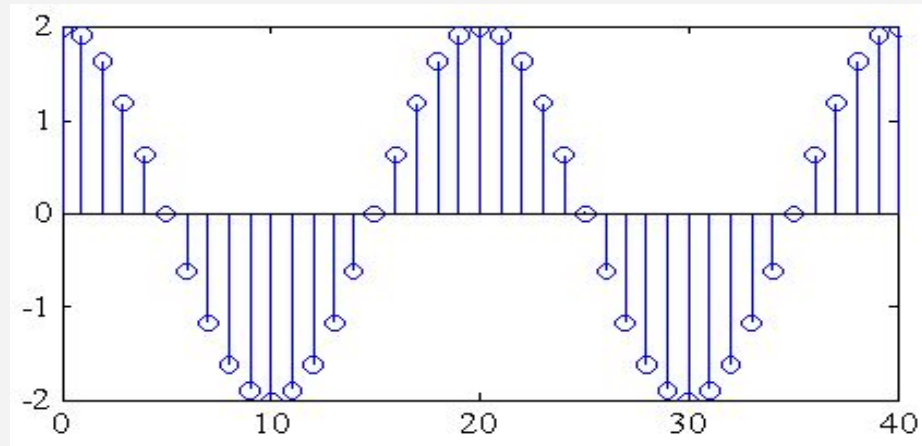
Basic Sequences

- Real sinusoidal sequence -

$$x[n] = A \cos(\omega_0 n + \varphi)$$

where A is the amplitude, ω_0 is the angular frequency, and φ is the phase of $x[n]$

Example -



Basic Sequences

- Exponential sequence -

$$x[n] = A\alpha^n, \quad -\infty < n < \infty$$

where A and α are real or complex numbers

If we write : $\alpha = e^{(\sigma_o + j\omega_o)}$, $A = |A|e^{j\phi}$,

then we can express

$$x[n] = |A|e^{j\phi}e^{(\sigma_o + j\omega_o)n} = x_{re}[n] + jx_{im}[n],$$

where $x_{re}[n] = |A|e^{\sigma_o n} \cos(\omega_o n + \phi),$

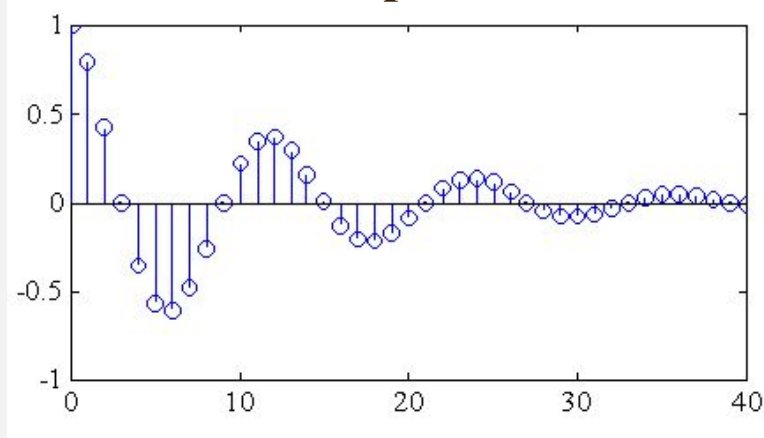
$$x_{im}[n] = |A|e^{\sigma_o n} \sin(\omega_o n + \phi)$$



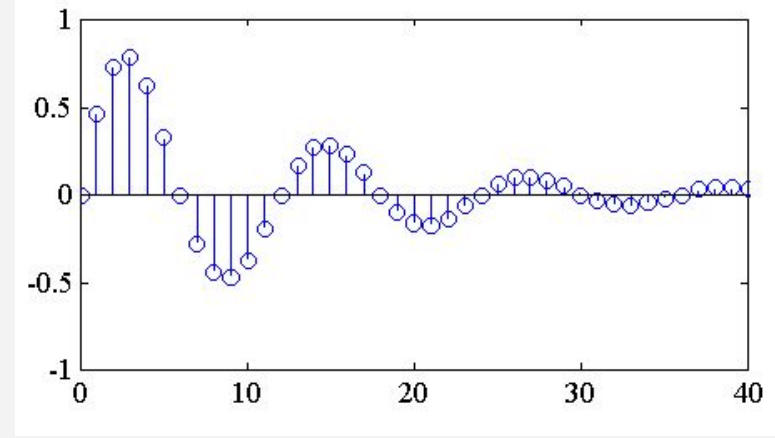
Basic Sequences

- $x_{\text{re}}[n]$ and $x_{\text{im}}[n]$ of a complex exponential sequence are real sinusoidal sequences with constant ($\sigma_0=0$), growing ($\sigma_0>0$), and decaying ($\sigma_0<0$) amplitudes for $n > 0$

Real part



Imaginary part



$$x[n] = \exp\left(-\frac{1}{12} + j\frac{\pi}{6}\right)n$$



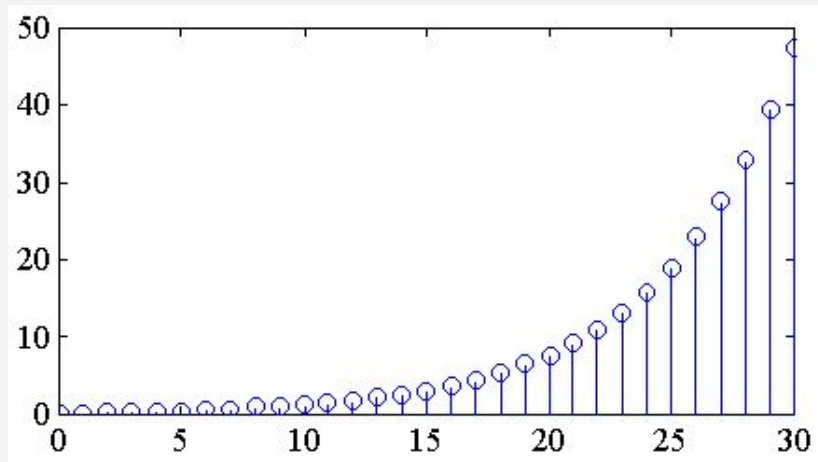
Basic Sequences

- Real exponential sequence -

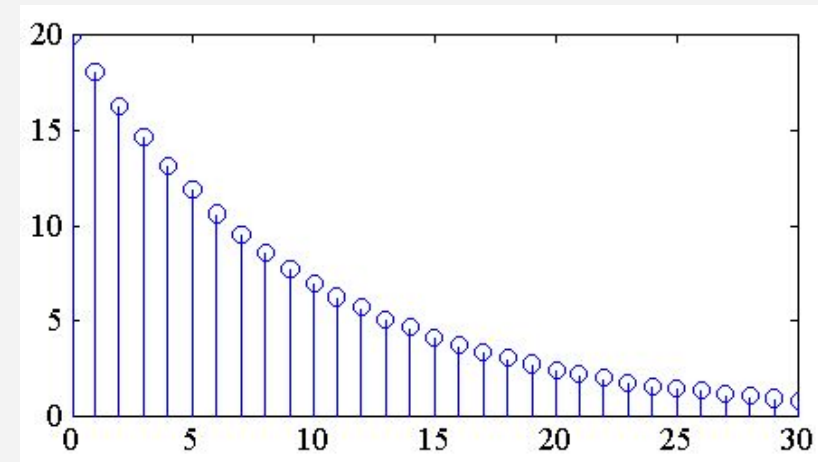
$$x[n] = A\alpha^n, \quad -\infty < n < \infty$$

where A and α are real numbers

$\alpha=1.2$



$\alpha=0.9$



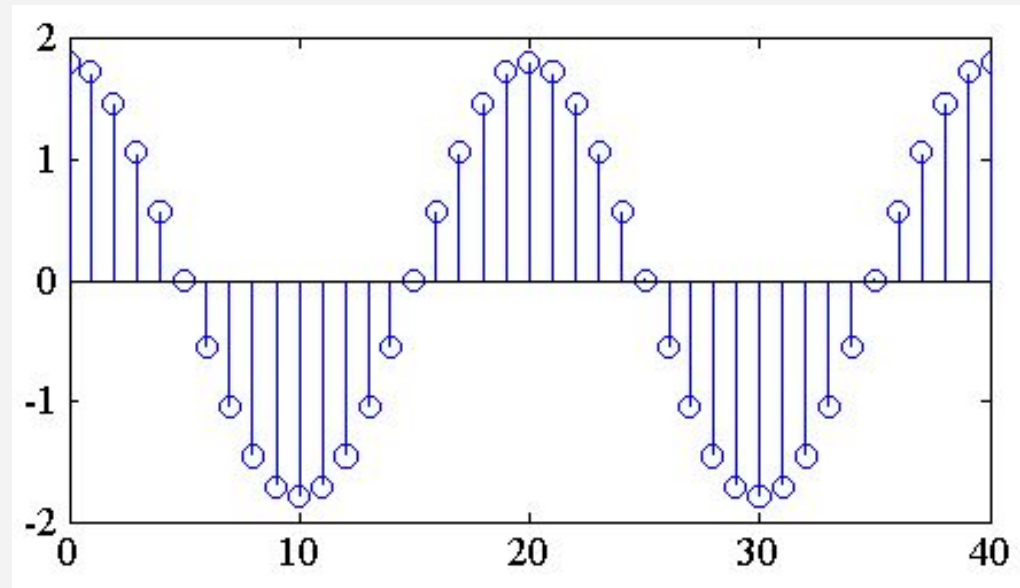
Basic Sequences

- Sinusoidal sequence $A\cos(\omega_0 n + \varphi)$ and complex exponential sequence $Be^{j\omega_0 n}$ are periodic sequences of period N if $\omega_0 N = 2\pi r$ where N and r are positive integers
- Smallest value of N satisfying $\omega_0 N = 2\pi r$ is the fundamental period of the sequence



Basic Sequences

$$\omega_0 = 0.1\pi$$

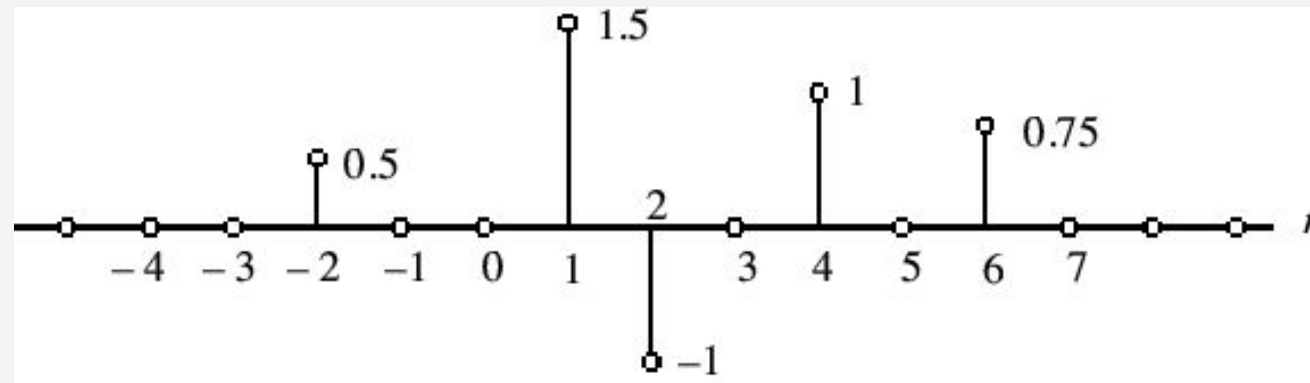


- Here $\omega_0 = 0.1\pi$
- Hence $N = 2\pi r / \omega_0 = 20$ for $r = 1$



Basic Sequences

- An arbitrary sequence can be represented in the time-domain as a weighted sum of some basic sequence and its delayed (advanced) versions

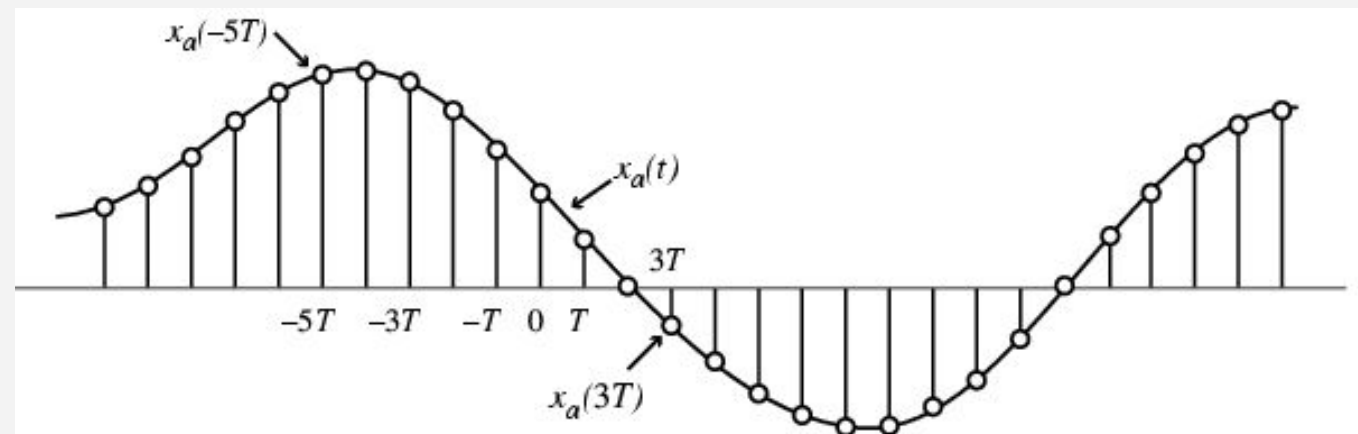


$$x[n] = 0.5\delta[n+2] + 1.5\delta[n-1] - \delta[n-2] + \delta[n-4] + 0.75\delta[n-6]$$



The Sampling Process

- Often, a discrete-time sequence $x[n]$ is developed by uniformly sampling a continuous-time signal $x_a(t)$ as indicated below



- The relation between the two signals is
$$x[n] = x_a(t)|_{t=nT} = x_a(nT), \quad n = \dots, -2, -1, 0, 1, 2, \dots$$



The Sampling Process

- Time variable t of $x_a(t)$ is related to the time variable n of $x[n]$ only at discrete-time instants t_n given by

$$t_n = nT = n/F_T = 2\pi n / \Omega_T$$

with $F_T = 1/T$ denoting the **sampling frequency** and $\Omega_T = 2\pi F_T$ denoting the **sampling angular frequency**



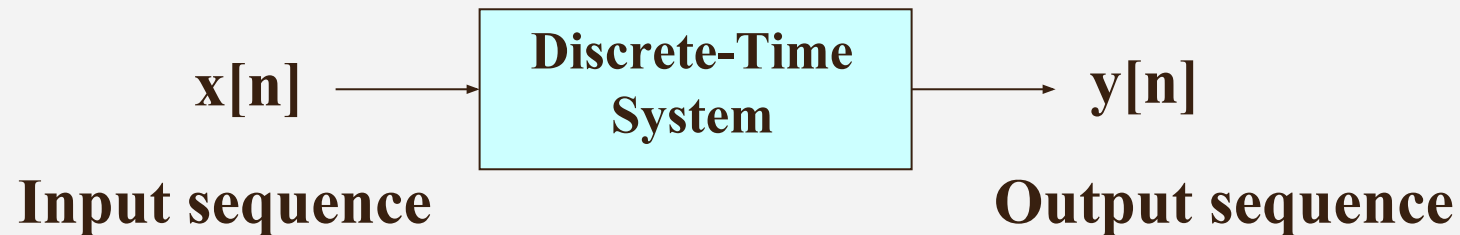
The Sampling Process

- **Generalization: Consider an arbitrary continuous-time signal $x_a(t)$ composed of a weighted sum of a number of sinusoidal signals**
- **$x_a(t)$ can be represented uniquely by its sampled version $\{x[n]\}$ if the sampling frequency Ω_T is chosen to be greater than 2 times the highest frequency contained in $x_a(t)$**
- **The condition to be satisfied by the sampling frequency to prevent aliasing is called the **sampling theorem (Nyquist theorem)****



Discrete-Time Systems

- A discrete-time system processes a given input sequence $x[n]$ to generate an output sequence $y[n]$ with more desirable properties
- In most applications, the discrete-time system is a single-input, single-output system:



Discrete-Time Systems

- **Linear System**
- **Shift-Invariant System**
- **Causal System**
- **Stable System**
- **Passive and Lossless Systems**



Linear Discrete-Time Systems

- **Definition** - If $y_1[n]$ is the output due to an input $x_1[n]$ and $y_2[n]$ is the output due to an input $x_2[n]$ then for an input

$$x[n] = ax_1[n] + bx_2[n]$$

the output is given by

$$y[n] = ay_1[n] + by_2[n]$$

- Above property must hold for any arbitrary constants a and b and for all possible inputs $x_1[n]$ and $x_2[n]$
- Hence, the above system is **linear**



Shift-Invariant System

- For a shift-invariant system, if $y_1[n]$ is the response to an input $x_1[n]$, then the response to an input

$$x[n]=x_1[n-n_0]$$

is simply

$$y[n]=y_1[n-n_0]$$

where n_0 is any positive or negative integer

- The above relation must hold for any arbitrary input and its corresponding output
- The above property is called **time-invariant** property, or **shift-invariant** property



Linear Time-Invariant system

- **Linear Time-Invariant (LTI) System -**

A system satisfying both the linearity and the time-invariant property

- LTI systems are mathematically easy to analyze and characterize, and consequently, easy to design
- Highly useful signal processing algorithms have been developed utilizing this class of systems over the last several decades



Impulse and Step Responses

- The response of a discrete-time system to a unit sample sequence $\delta[n]$ is called the **unit impulse response** or simply, the **impulse response**, and is denoted by $h[n]$
- The response of a discrete-time system to a unit step sequence $\mu[n]$ is called the **unit step response** or simply, the **step response**, and is denoted by $s[n]$



Time-Domain Characterization of LTI Discrete-Time Systems

- Input-Output Relationship - A consequence of the linear, time-invariance property is that an **LTI discrete-time system is completely characterized by its impulse response**
- Knowing the impulse response one can compute the output of the system for any arbitrary input



Time-Domain Characterization of LTI Discrete-Time Systems

- Now, any arbitrary input sequence $x[n]$ can be expressed as a linear combination of delayed and advanced unit sample sequences in the form

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

- The response of the LTI system to an input $x[k] \delta[n-k]$ will be $x[k] h[n-k]$



Time-Domain Characterization of LTI Discrete-Time Systems

- Hence, the response $y[n]$ to an input

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

will be $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

which can be alternately written as

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k] h[k]$$



Convolution Sum

- The summation

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k] h[k]$$

is called the **convolution sum** of the sequences $x[n]$ and $h[n]$ and represented compactly as

$$y[n] = x[n] \circledast h[n]$$



Convolution Sum

Properties -

▪ **Commutative property:**

$$x[n] \otimes h[n] = h[n] \otimes x[n]$$

• **Associative property :**

$$(x[n] \otimes h[n]) \otimes y[n] = x[n] \otimes (h[n] \otimes y[n])$$

• **Distributive property :**

$$x[n] \otimes (h[n] + y[n]) = x[n] \otimes h[n] + x[n] \otimes y[n]$$



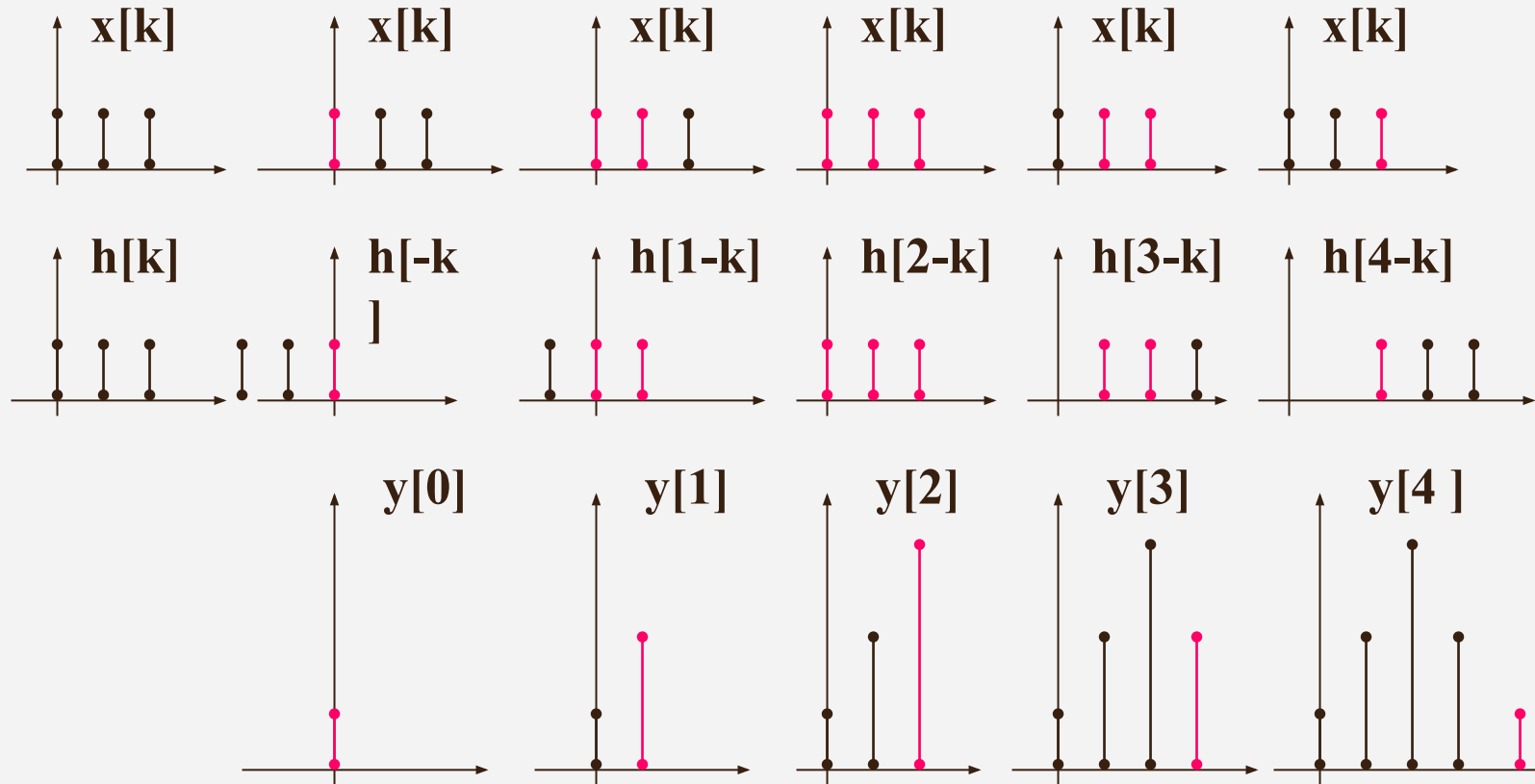
Convolution Sum

Interpretation -

- 1) Time-reverse $h[k]$ to form $h[-k]$
- 2) Shift $h[-k]$ to the right by n sampling periods if $n > 0$ or shift to the left by n sampling periods if $n < 0$ to form $h[n-k]$
- 3) Form the product $v[k]=x[k]h[n-k]$
- 4) Sum all samples of $v[k]$ to develop the n -th sample of $y[n]$ of the convolution sum



$$x[n] = h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$



$$y[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 2\delta[n-3] + \delta[n-4]$$



Convolution Sum

In general, if the lengths of the two sequences being convolved are M and N , then the sequence generated by the convolution is of length **$M+N-1$**



Classification of LTI Discrete-Time Systems

Based on Impulse Response Length -

- If the impulse response $h[n]$ is of finite length, i.e.,

$$h[n]=0 \quad \text{for } n < N_1, N_2 < n \text{ and } N_1 < N_2$$

then it is known as a **Finite Impulse Response (FIR)** discrete-time system

- The convolution sum description here is

$$y[n] = \sum_{k=N_1}^{N_2} h[k]x[n-k]$$



Classification of LTI Discrete-Time Systems

The output $y[n]$ of an FIR LTI discrete-time system can be computed directly from the convolution sum as it is a finite sum of products



Classification of LTI Discrete-Time Systems

- If the impulse response is of infinite length, then it is known as an **Infinite Impulse Response (IIR)** discrete-time system
- The class of IIR systems we are concerned with in this course are characterized by linear constant coefficient difference equations



Classification of LTI Discrete-Time Systems

- Example - The discrete-time accumulator defined by

$$y[n] = y[n-1] + x[n]$$

is seen to be an IIR system



Classification of LTI Discrete-Time Systems

Based on the Output Calculation Process

- **Non-recursive System** - Here the output can be calculated sequentially, knowing only the present and past input samples
- **Recursive System** - Here the output computation involves past output samples in addition to the present and past input samples



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